

ArogyaRakshak: A Multi-Agent AI Decision-Support Platform for Hospital Bill Auditing, Insurance Claim Assistance, Government Scheme Discovery, and Medicine Price Verification

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Domain Keywords: Healthcare Decision Support, Multi-Agent LLM Systems, Retrieval-Augmented Generation (RAG), Optical Character Recognition (OCR), Indic / Cross-Lingual NLP, Health Insurance Regulation (IRDAI), Government Welfare Schemes (PMJAY/MJPJAY), FastAPI, Next.js

Background / Importance of Topic

Patients discharged from Indian hospitals routinely face four compounding sources of financial and administrative friction: (i) itemised hospital bills that overcharge against published Central Government Health Scheme (CGHS) benchmark rates; (ii) health-insurance claims that are wrongfully repudiated by citing clauses such as the pre-existing-condition moratorium even when regulatory conditions for such denial are not met; (iii) low awareness of entitlements under national and state welfare schemes such as PMJAY and MJPJAY; and (iv) branded medicines purchased at retail MRP when NPPA price-controlled generic equivalents exist at Jan Aushadhi Kendras for a fraction of the cost. Each of these problems has historically been addressed, if at all, by isolated tools or manual advocacy. There is a need for a single, patient-facing platform that can read a patient's own documents once and surface actionable guidance across all four problem areas, in the patient's own language.

Purpose / Problem Statement

To design and build a unified, multilingual (English, Hindi, Marathi) decision-support platform that ingests a patient's own healthcare documents (hospital bill, prescription, insurance denial letter, or policy) and, without requiring the document to be re-entered for each concern, produces: (a) a line-item hospital bill audit against CGHS rates with a dispute letter; (b) a pre-filled insurance claim / pre-authorisation package; (c) a regulatory audit of claim denials with a drafted multi-tier statutory appeal; (d) a government welfare scheme eligibility result; and (e) a medicine price/generic-substitute check — while keeping the uploaded document itself in transient memory only, never on persistent storage.

System / Algorithmic Approach

1. Document Intake (BYOD): The patient uploads a bill, prescription, denial letter, or policy document through the web (Next.js) or mobile (React Native/Expo) client. The raw file is held only in a transient in-memory buffer for the duration of the request (Bring-Your-Own-Document, zero persistent storage).
2. Multilingual OCR: The shared Kadi layer runs OCR (EasyOCR / PyMuPDF) capable of reading Devanagari and Latin-script text from scanned images and PDFs.
3. Entity Extraction & Normalisation: Extracted text is passed to a Groq-hosted LLM (openai/gpt-oss-120b), with a deterministic regex/heuristic fallback, to normalise patient, hospital, diagnosis, procedure, medicine, and billing line-item entities into a shared JSON schema, persisted against a consent-gated case session.
4. Cross-Module Context Sharing: Once a patient opts in, the same extracted entities are made available to every downstream module via Kadi's FAISS-backed in-memory vector store, avoiding repeated uploads. Kadi's design additionally specifies phonetic transliteration (IndicXlit) and cross-lingual semantic embedding (IndicSBERT) scoring to resolve the same entity written in different scripts or languages; this combined resolution pipeline is the platform's next planned development milestone and is not yet wired into the running system.
5. Domain-Specific Multi-Agent / RAG Reasoning: The relevant module runs its own reasoning chain grounded in a public government reference source — BillNyay's 5-agent chain (Document Auditor → Clinical Reviewer → Regulatory Advisor → Appeal Drafter → QA Judge) audits bills against CGHS rates; BimaNyay audits denial letters against the IRDAI Master Circular (29 May 2024); SchemeSetu reasons over PMJAY/MJPJAY eligibility rules; DawaCheck benchmarks

MRP against NPPA Schedule-I ceiling prices; DaaviSetu assembles pre-authorisation/claim packages.

6. Real-Time Progress Streaming: A FastAPI Server-Sent Events (SSE) channel streams pipeline stage progress (document received → OCR → extraction → domain audit → completed) to the client during processing.
7. Output Generation: The system returns a structured audit report, a drafted appeal/dispute letter (PDF, via ReportLab), a claim/pre-authorisation form, an eligibility verdict with a step-by-step claim guide, or a price/generic comparison, depending on the module invoked — rendered trilingually where the module supports it.

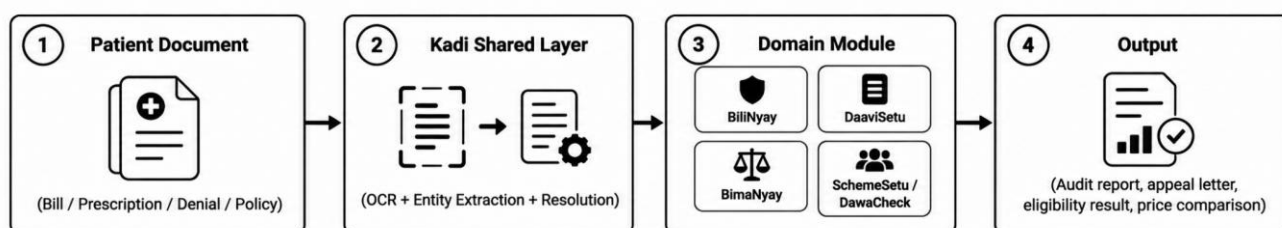


Figure 1: Compact System Pipeline

Reference Data & Inputs

Government Reference Datasets: CGHS rate schedules (BillNyay); NPPA Schedule-I ceiling price orders (DawaCheck); PMJAY (national) and MJPJAY (Maharashtra) scheme rules (SchemeSetu); IRDAI Master Circular, 29 May 2024, and Insurance Ombudsman Rules, 2017 (BimaNyay)

User Inputs: Uploaded hospital bill / prescription / denial letter / policy document (image or PDF); short structured intake forms (income, category, diagnosis, medicine name) where a document upload is not required

System Outputs: Line-item overcharge audit and dispute letter; pre-authorisation/claim package; IRDAI violation audit with 3-tier appeal drafts (GRO, Bima Bharosa, Ombudsman Form VI) and statutory SLA countdown; scheme eligibility verdict and claim guide; medicine MRP overcharge flag and generic substitute list

Major Modules & Key Technologies

- Kadi (shared layer) — OCR, entity extraction/normalisation, FAISS vector store, consent-gated context sharing
- BillNyay — 5-agent hospital bill audit vs. CGHS rates
- DaaviSetu — pre-claim cashless pre-authorisation / reimbursement form automation
- BimaNyay — post-denial IRDAI regulatory audit and 3-tier statutory appeal drafting with SLA tracking
- SchemeSetu — PMJAY/MJPJAY eligibility reasoning with an offline embedding fallback
- DawaCheck — NPPA Schedule-I price verification and Jan Aushadhi generic substitution
- Platform: FastAPI (async) · PostgreSQL 16 + pgvector · FAISS · Groq Cloud LLM inference · Next.js 16 / React 19 web client · React Native + Expo mobile client · Docker Compose

Conclusions / Applications

- Replaces five fragmented, manual advocacy processes (bill audit, claim filing, denial appeal, scheme discovery, medicine price check) with one document upload.
- Grounds every generated claim in a citable, public government reference source rather than an unverified model output.
- Serves patients, caregivers, and hospital billing desks in English, Hindi, and Marathi across web and mobile.
- Current build (Product Phase 1) has all five modules wired end-to-end on both clients against real backend endpoints, verified by an automated test suite; combined phonetic + semantic cross-lingual entity resolution in Kadi is the next planned milestone (Product Phase 2), not yet part of the running pipeline.

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Signature of Guide: _____

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